Deep Unfolded Fractional Programming-Based Beamforming in RIS-Aided MISO Systems

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***Abstract*—Reconfigurable intelligent surface (RIS) is a promising solution to programmable wireless channels. However, most existing RIS phase optimization algorithms, relying on iterative processes, suffer from high latency and poor scalability. To address this issue, we propose a low-complexity scheme based on deep unfolding. Specifically, we consider a downlink MISO system aided by a RIS and aim to maximize users' weighted sumrate (WSR). We employ Lagrangian dual transformation to decouple the original non-convex problem into two subproblems: transmit beamforming optimization and phase shift design. Then, we introduce a block coordinate descent (BCD) method, which still relies on iterative updates and includes complex operations such as matrix inversion, leading to high computational complexity and latency. To achieve fast solutions, we further propose to unfold the BCD method's iterative process into layers of an interpretable neural network (NN) with a few trainable parameters. The NN is trained offline and deployed online for realtime solutions. Finally, numerical results validate the performance of the proposed scheme in terms of comparable WSR and reduced computational complexity.**

***Index Terms*—RIS, beamforming, fractional programming, deep unfolding.**

I. Introduction

NEW PHYSICAL layer techniques are being explored to meet the high data rate, reliability, and energy efficiency requirements of the upcoming 6G wireless communication systems. Among these techniques, Reconfigurable intelligent surface (RIS), which is composed of meta-material reflective units, is considered to have potential to replace traditional relays for wireless communication assistance due to its low energy consumption, low cost, and simple deployment. For example, the most direct application scenario is to help signals bypass the obstacles through the new links generated by RIS, so as to improve the quality of service of networks [1]. RIS can also be used to enhance the received signal strength of users at network edge or eliminate the impact of inter-cell interference. In addition, RIS can support large-scale connectivity by programming wireless channels [2].

There have been many works studying RIS-aided communication systems, e.g., [3], [4], [5], [6]. In these works, phase shift of RIS elements and transmit/receive beamforming are jointly optimized using iterative process, which is not scalable to large-scale RIS-assisted networks due to the huge number of involved variables. Consequently, the computational complexity caused by iterative process makes the joint optimization algorithms have difficulty in satisfying real-time demands especially for delay-sensitive applications.

In recent years, deep learning (DL) technique is widely exploited in physical layer issues, which trains neural networks offline and do inference online with some simple operations [7]. The work [8] proposed a DL based approach for RIS optimization by choosing discrete phase vectors from a pre-defined codebook whereas the work [9] used a deep neural network (NN) to predict continuous phase values of RIS directly. The authors in [10] further proposed to predict RIS phase and beamforming vectors directly at the same time using a deep NN. However, black-box deep NNs are data-driven and require a huge number of data samples for training, which may not be easy to implement in some cases with scarce data samples. Black-box deep NNs also have poor interpretability and generalization ability. Besides, the number of trainable parameters increases significantly as network scale expands, leading to higher prediction complexity and longer training time. To overcome the drawbacks, deep unfolding method is developed to balance system performance and complexity by integrating domain knowledge of signal processing into NNs [11]. For example, considering the complexity of scalability of the iterative weighted minimum mean square error (WMMSE) algorithm, [12], [13] proposed to unfold it into layers of a NN with reduced trainable parameters, thereby accelerating calculation process. The unfolded WMMSE algorithm was extended to the multi-cell scenario in [14]. Authors in [15] unfolded the structure of projection onto convex set with bounded perturbation to find fast solutions to multicast beamforming problem. Besides, deep unfolding NNs based on the alternating direction method of multipliers (ADMM) algorithm were applied to compressed sensing [16] and beamforming design [17].

In this letter, we investigate the joint optimization problem of transmit beamforming and RIS phase shift in a multiple-input single-output (MISO) downlink transmission system aided by a multi-element RIS, aiming to maximize the weighted sumrate (WSR) of users. Considering the multi-ratio fractional programming (FP) structure of the objective function, we decouple the non-trivial problem into two subproblems, i.e., the beamforming optimization of base station (BS) and the phase optimization of RIS, through Lagrange dual transformation. Then, a block coordinate descent (BCD)

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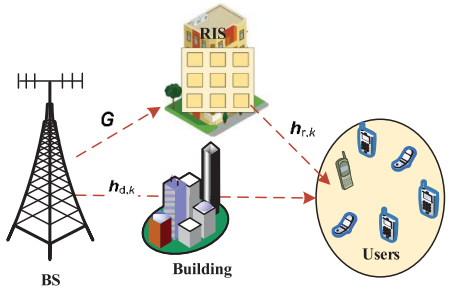


Fig. 1. The snapshot of the considered system.

method is introduced, which still relies on iterative process and includes complex operations such as matrix inversion and Armijo search. The main contributions of this letter are summarized as follows.

- To further reduce computational complexity and latency, this letter proposes a deep unfolding FP (DUFP) scheme which unfolds the iterative process of the BCD method into several layers of a NN. Different from the conventional black-block NNs, the NN is interpretable because the structure of each layer is self-defined according to domain knowledge of signal processing.
- In addition, the proposed scheme is scalable because the number of trainable parameters (i.e., the step sizes) is independent of network size and the step sizes are shared when updating the transmit beamforming and RIS phase shift.
- Finally, numerical results validate that the proposed scheme can achieve comparable WSR performance with reduced computational complexity.

II. System Model

As shown in Fig. 1, we consider a RIS-assisted downlink MISO system, where a BS equipped with M antennas serves K single-antenna users. Considering that the direct links between the BS and users may suffer from deep fading due to the obstruction of tall buildings, and thus a RIS equipped with N reflective elements is deployed on the facade of surrounding buildings to provide spatial diversity and help the BS overcome adverse propagation conditions. Define $\mathbf{h}_{d,k} \in \mathbb{C}^{M\times 1}$, $\mathbf{h}_{r,k} \in \mathbb{C}^{N\times 1}$, and $\mathbf{G} \in \mathbb{C}^{N\times M}$ denote the channel between the BS and user k, the channel between the RIS and user k, and the channel between the BS and RIS, respectively. The phase shift vector of the RIS is defined as $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_N]^T$, where θ_n is the phase shift of the n-th reflection element of the RIS.

The signal received by user k is represented as

$$y_k = \left(\mathbf{h}_{d,k}^{\mathrm{H}} + \mathbf{h}_{r,k}^{\mathrm{H}} \operatorname{diag}(\boldsymbol{\theta})\mathbf{G}\right) \sum_{k=1}^{K} \mathbf{v}_k s_k + n_k, \tag{1}$$

where s_k the data symbol for user k, $\mathbf{v}_k \in \mathbb{C}^{M\times 1}$ is the beamforming vector of user k, and $n_k \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise. Here, $\mathcal{CN}(0, \sigma^2)$ denotes the circularly symmetric complex Gaussian distribution with zero mean and variance σ^2. Then, the signal-to-interference-plus-noise ratio (SINR) of user k is represented as

$$\alpha_k = \frac{\left|\left(\mathbf{h}_{d,k}^{\mathrm{H}} + \mathbf{h}_{r,k}^{\mathrm{H}} \operatorname{diag}(\boldsymbol{\theta})\mathbf{G}\right)\mathbf{v}_k\right|^2}{\sum_{i=1,i\neq k}^{K}\left|\left(\mathbf{h}_{d,k}^{\mathrm{H}} + \mathbf{h}_{r,k}^{\mathrm{H}} \operatorname{diag}(\boldsymbol{\theta})\mathbf{G}\right)\mathbf{v}_i\right|^2 + \sigma_0^2}. \tag{2}$$

The goal of this letter is to maximize the WSR of the users by jointly optimizing the transmit beamforming of the BS and the phase shift of the RIS. The WSR maximization problem is formulated as

$$(\mathcal{P}_1): \max_{\mathbf{V},\boldsymbol{\theta}} \sum_{k=1}^{K} \omega_k \log(1+\alpha_k)$$
$$\text{s.t.} \quad |\theta_n| = 1, \forall n, \tag{3}$$
$$\sum_{k=1}^{K} \|\mathbf{v}_k\|^2 \le P_{\max}, \tag{4}$$

where $P_{\max}$ is the power budget of the BS, $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_K]$, and ω_k is the weight coefficient of user k.

Problem $(\mathcal{P}_1)$ is a typical multi-ratio FP problem. In the following, we first introduce an iterative algorithm named BCD method based on Lagrange dual transformation and then propose the DUFP scheme.

III. Block Coordinate Descent Method

The Lagrange dual transformation is adopted to transform the objective function of problem $(\mathcal{P}_1)$ into a more tractable form [18], i.e.,

$$(\mathcal{P}_{1\text{-}1}): \max_{\mathbf{V},\boldsymbol{\theta},\boldsymbol{\gamma}} \sum_{k=1}^{K} \omega_k \left[\log(1+\gamma_k) - \gamma_k + \frac{(1+\gamma_k)\alpha_k}{1+\alpha_k}\right]$$
$$\text{s.t.} \quad (3) \text{ and } (4),$$

where $\boldsymbol{\gamma} = \{\gamma_k, \forall k\}$ is a collection of auxiliary variables whose optimal solution is $\gamma_k^* = \alpha_k, \forall k$. According to [18, Th. 3], problem $(\mathcal{P}_{1\text{-}1})$ is equivalent to problem $(\mathcal{P}_1)$. In other words, $\{\boldsymbol{\theta}, \boldsymbol{\gamma}\}$ is the solution to problem $(\mathcal{P}_{1\text{-}1})$ if and only if it is the solution to problem $(\mathcal{P}_1)$, and the optimal objective values of the two problems are also equal. Thus, when $\boldsymbol{\gamma}$ is fixed, problem $(\mathcal{P}_1)$ can be rewritten as

$$(\mathcal{P}_2): \max_{\mathbf{V},\boldsymbol{\theta}} \sum_{k=1}^{K} \frac{\omega_k(1+\gamma_k)\alpha_k}{1+\alpha_k}$$
$$\text{s.t.} \quad (3) \text{ and } (4).$$

In the following, we will give the update rules of $\mathbf{V}$ and $\boldsymbol{\theta}$.

A. Update of V

Define $\mathbf{h}_k = \mathbf{h}_{d,k} + \mathbf{G}^H \operatorname{diag}(\boldsymbol{\theta})\mathbf{h}_{r,k}$. By substituting (2) into the objective function of problem $(\mathcal{P}_2)$, we have

$$f_1(\mathbf{V}) = \sum_{k=1}^{K} \frac{\omega_k(1+\gamma_k)\left|\mathbf{h}_k^H \mathbf{v}_k\right|^2}{\sum_{i=1}^{K}\left|\mathbf{h}_k^H \mathbf{v}_i\right|^2 + \sigma_0^2}, \tag{5}$$

which is a function of $\mathbf{V}$. Therefore, when $\boldsymbol{\gamma}$ and $\boldsymbol{\theta}$ are given, the problem of optimizing $\mathbf{V}$ is ultimately expressed as

$$(\mathcal{P}_3): \max_{\mathbf{V}} \quad f_1(\mathbf{V})$$
$$\text{s.t.} \quad (4).$$

Note that problem $(\mathcal{P}_3)$ is still a multi-ratio FP problem and the quadratic transformation in [18] can be adopted. Then, $f_1(\mathbf{V})$ is reconstructed as

$$f_1'(\mathbf{V}) = \sum_{k=1}^{K} 2\sqrt{\omega_k(1+\gamma_k)}\operatorname{Re}\left\{z_k^* \mathbf{h}_k^H \mathbf{v}_k\right\}$$
$$- \sum_{k=1}^{K} |z_k|^2 \left(\sum_{i=1}^{K}\left|\mathbf{h}_k^H \mathbf{v}_i\right|^2 + \sigma_0^2\right), \tag{6}$$

where $z_k \in \mathbb{C}$ is an auxiliary variable. Therefore, problem $(\mathcal{P}_3)$ is equivalent to

$$(\mathcal{P}_4): \max_{\mathbf{V},\mathbf{z}} \quad f_1'(\mathbf{V}) \quad \text{s.t.} \quad (4).$$

where $\mathbf{z} = [z_1, \ldots, z_K]^T$. Problem $(\mathcal{P}_4)$ is a nonconvex optimization problem with respective to $(\mathbf{V}, \mathbf{z})$, but it is convex with respect to $\mathbf{V}$ and $\mathbf{z}$, respectively. A common solution to such a problem is to alternately update $\mathbf{V}$ and $\mathbf{z}$ by fixing the other one. Specifically, z_k is found via $\partial f_1'(\mathbf{V})/\partial z_k = 0$, i.e.,

$$z_k^{\circ} = \frac{\sqrt{\omega_k(1+\gamma_k)}\mathbf{h}_k^{\mathrm{H}}\mathbf{v}_k}{\sum_{i=1, i\neq k}^{K} \left|\mathbf{h}_k^{\mathrm{H}}\mathbf{v}_i\right|^2 + \sigma_0^2}. \tag{7}$$

Similarly, the update rule of $\mathbf{v}_k$ is given as

$$\mathbf{v}_k^{\circ} = \sqrt{\omega_k(1+\gamma_k)} z_k (\eta \mathbf{I}_M + \mathbf{A})^{-1}\mathbf{h}_k, \tag{8}$$

where $\mathbf{I}_M$ is the identity matrix of dimension M, $\mathbf{A} = \sum_{i=1}^{K} |z_i|^2 \mathbf{h}_i \mathbf{h}_i^{\mathrm{H}}$, and η is the Lagrange multiplier associated with the power constraint, given as

$$\eta^{\circ} = \min\left\{ \eta \geq 0 : \sum_{k=1}^{K} \|\mathbf{v}_k(\eta)\|_2^2 \leq P_{\max} \right\}. \tag{9}$$

Note that an iterative process involving matrix inversion operation is required to the solution of η. In general, a higher-precision $\mathbf{V}$ requires more iterations, which leads to higher computational complexity.

B. Update of $\boldsymbol{\theta}$

Define $\mathbf{a}_{i,k} = \mathrm{diag}(\mathbf{h}_{r,k}^{\mathrm{H}})\mathbf{G}\mathbf{v}_i$ and $b_{i,k} = \mathbf{h}_{d,k}^{\mathrm{H}}\mathbf{v}_i$. The objective function of problem $(\mathcal{P}_2)$ can be rewritten as

$$f_2(\boldsymbol{\theta}) = \sum_{k=1}^{K} \frac{\omega_k(1+\gamma_k)\left|b_{k,k} + \boldsymbol{\theta}^{\mathrm{H}}\mathbf{a}_{k,k}\right|^2}{\sum_{i=1}^{K}\left|b_{i,k} + \boldsymbol{\theta}^{\mathrm{H}}\mathbf{a}_{i,k}\right|^2 + \sigma_0^2}. \tag{10}$$

The quadratic transformation is adopted again, then we have

$$f_2'(\boldsymbol{\theta}) = \sum_{k=1}^{K} 2\sqrt{\omega_k(1+\gamma_k)}\,\mathrm{Re}\left\{ z_k^* \boldsymbol{\theta}^H \mathbf{a}_{k,k} + z_k^* b_{k,k} \right\} - \sum_{k=1}^{K} |z_k|^2 \left(\sum_{i=1}^{K} \left| b_{i,k} + \boldsymbol{\theta}^H \mathbf{a}_{i,k} \right|^2 + \sigma_0^2 \right), \tag{11}$$

where z_k is updated using (7). Then, the objective function of the phase optimization problem can be ultimately expressed as

$$f_3(\boldsymbol{\theta}) = -\boldsymbol{\theta}^H \mathbf{U} \boldsymbol{\theta} + 2\,\mathrm{Re}\left\{ \boldsymbol{\theta}^H \boldsymbol{\nu} \right\} + \chi, \tag{12}$$

where χ is a term unrelated to $\boldsymbol{\theta}$,

$$\mathbf{U} = \sum_{k=1}^{K} |z_k|^2 \sum_{i=1}^{K} \mathbf{a}_{i,k}\mathbf{a}_{i,k}^{\mathrm{H}}, \tag{13}$$

and

$$\boldsymbol{\nu} = \sum_{k=1}^{K} \left(\sqrt{\omega_k(1+\gamma_k)} z_k^* \mathbf{a}_{k,k} - |z_k|^2 \sum_{i=1}^{K} b_{i,k}^* \mathbf{a}_{i,k} \right). \tag{14}$$

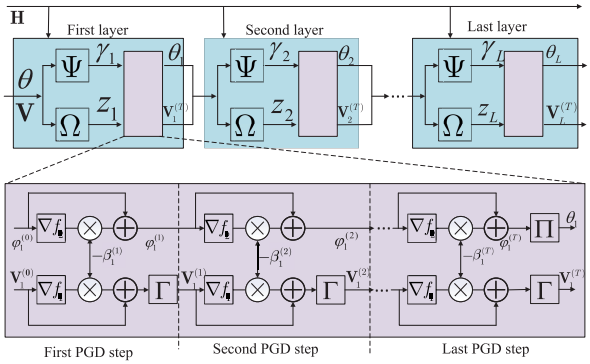


Fig. 2. The NN structure of the proposed DUFP scheme. The operators Ψ and Ω denotes the update operations of $\boldsymbol{\gamma}$ (i.e., $\gamma_k = \alpha_k, \forall k$) and $\mathbf{z}$ (i.e., (7)), respectively.

Therefore, the update rule of the phase shift vector $\boldsymbol{\theta}$ is given as follows

$$\boldsymbol{\theta} = \arg\min_{\boldsymbol{\theta}} f_3'(\boldsymbol{\theta}) \triangleq \boldsymbol{\theta}^H \boldsymbol{U} \boldsymbol{\theta} - 2\,\mathrm{Re}\left\{ \boldsymbol{\theta}^H \boldsymbol{\nu} \right\}. \tag{15}$$

Due to the constraint in (3), we substitute $\theta_n = e^{\jmath\varphi_n}$ into (15) and then have

$$\boldsymbol{\varphi} = \arg\min_{\boldsymbol{\varphi}\in\mathbb{R}^N} f'(\boldsymbol{\varphi}) \triangleq (e^{\jmath\boldsymbol{\varphi}})^H \boldsymbol{U} e^{\jmath\boldsymbol{\varphi}} - 2\mathrm{Re}\left\{ \boldsymbol{\nu}^H e^{\jmath\boldsymbol{\varphi}} \right\}, \tag{16}$$

which is a nonconvex problem whose optimal solution is hard find. According to [19], we can resort to successive convex approximation (SCA) technique.

Then, a BCD method based on the Lagrangian dual transform is proposed, which update $\mathbf{z}$, $\mathbf{V}$, $\boldsymbol{\varphi}$, and $\boldsymbol{\alpha}$ repeatedly until convergence. Note that in each outer iteration, two inner iteration processes are also required to find η satisfying (9) and $\boldsymbol{\varphi}$ using the SCA technique. The BCD method is a two-layer iterative process which includes complex operations such as matrix inversion and one dimensional search [3], leading to high computational complexity and latency. To deal with this issue, we proposed the DUFP scheme.

IV. Proposed Deep Unfolding Based Scheme

The NN structure of the proposed DUFP scheme is shown in Fig. 2, which includes L layers each with T projected gradient descent (PGD) steps. We use the subscript $(\cdot)_l$ and the superscript $(\cdot)^{(t)}$ to denote the l-th layer and the t-th PGD step, respectively. Thus, $\mathbf{v}_{l,k}^{(t)}$ represents the t-th updating beamformer in the l-th iteration for user k. We also define $\nabla f_1(\mathbf{V}_l^{(t)}) = [\nabla f_1(\mathbf{v}_{l,1}^{(t)}) \ldots, \nabla f_1(\mathbf{v}_{l,K}^{(t)})]^T$, where $\nabla f_1(\mathbf{v}_{l,k}^{(t)}) = 2\sqrt{w_k(1+\gamma_k)} z_k \mathbf{h}_k - 2\mathbf{A}\mathbf{v}_{l,k}^{(t)}$.

In the lth layer, the t-th PGD update process is represented as

$$\begin{aligned} \mathbf{V}_l^{(t)} &= \mathbf{V}_l^{(t-1)} - \beta_l^{(t)} \nabla f_1\left(\mathbf{V}_l^{(t-1)}\right), \\ \mathbf{V}_l^{(t)} &= \Gamma\left\{\mathbf{V}_l^{(t)}\right\}. \end{aligned} \tag{17}$$

where $\beta_l^{(t)}$ is the step sizes and $\Gamma(\mathbf{V})$ is the normalization function defined as

$$\Gamma(\mathbf{V}) = \begin{cases} \mathbf{V}, & \text{if } \mathrm{Tr}\left(\mathbf{V}\mathbf{V}^H\right) \leq P_{\max}, \\ \dfrac{\mathbf{V}}{\|\mathbf{V}\|}\sqrt{P_{\max}}, & \text{otherwise.} \end{cases} \tag{18}$$

Similarly, the update rule of $\boldsymbol{\varphi}$ is given as

$$\boldsymbol{\varphi}_l^{(t)} = \boldsymbol{\varphi}_l^{(t-1)} - \beta_l^{(t)} \nabla f_2\left(\boldsymbol{\varphi}_l^{(t-1)}\right), \tag{19}$$

where $\nabla f_2(\boldsymbol{\varphi}) = 2\mathrm{Re}\{-\jmath\boldsymbol{\theta}^* \circ (\mathbf{U}\boldsymbol{\theta} - \boldsymbol{\nu})\}$. Note that the numerical solution to (16) can be obtained using finite PGD steps. Then, we use the function $\Pi(\boldsymbol{\varphi}) = e^{\jmath\boldsymbol{\varphi}}$ to restore $\boldsymbol{\theta}$.

We collect all the step sizes for each layer of PGD operations into a set $\boldsymbol{\beta} = [\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_L]$, where $\boldsymbol{\beta}_l = [\beta_l^{(1)}, \ldots, \beta_l^{(T)}]$. We denote each channel realization by $\mathbf{H} = \{\mathbf{G}, \mathbf{h}_{d,k}, \mathbf{h}_{r,k}, \forall k\}$. Then, the training loss function of the NN is defined as

$$\mathcal{L}_0 = \mathcal{L}_1 + \mathcal{L}_2,$$

where $\mathcal{L}_1$ and $\mathcal{L}_2$ represent the loss functions of the beamforming optimization module and the phase optimization module, respectively. Specifically, $\mathcal{L}_1$ is defined as

$$\mathcal{L}_1 = -\frac{1}{C}\sum_{c=1}^{C}\sum_{k=1}^{K} w_k \log\left(1 + \alpha_k\left(\mathbf{H}_{(c)}, \mathbf{V}_{(c)}\right)\right),$$

where $\mathbf{H}_{(c)}$ denotes the c-th channel realization, C is the number of channel realizations, $\mathbf{V}_{(c)}$ is the corresponding beamforming and updated according to (17), and α_k is calculated according to (2). In addition, $\mathcal{L}_2$ is defined as

$$\mathcal{L}_2 = \frac{1}{C}\sum_{c=1}^{C} f_3'\left(\mathbf{H}_{(c)}, \boldsymbol{\theta}_{(c)}\right),$$

where $f_3'(\mathbf{H}, \boldsymbol{\theta}) = \boldsymbol{\theta}^H \mathbf{U}\boldsymbol{\theta} - 2\,\mathrm{Re}\{\boldsymbol{\theta}^H \boldsymbol{\nu}\}$.

Note that the data flow of the proposed scheme is deterministic and the corresponding running time can be estimated, because the numbers of layers and PGD steps is determined in advance. In addition, to reduce the number of the trainable parameters, the learning rate matrix $\boldsymbol{\beta}$ is shared when updating $\mathbf{V}$ and $\boldsymbol{\theta}$. Consequently, the proposed scheme has TL trainable parameters, which is independent of the system size (i.e., K, M, and N) and thus is scalable. Finally, the computational complexity of updating $\mathbf{V}$ and $\boldsymbol{\theta}$ is $\mathcal{O}(L((T+1)KM^2 + TN^2 + K^2N^2))$.

V. Numerical Results

The simulation system consists of a BS equipped with $M = 6$ antennas, a RIS, and $K = 6$ users each with single antenna. These users are evenly and randomly distributed in a disc centered around (200m, 30m) with a radius of 10m. We assume that the channel between the BS and RIS, as well as the channel between RIS and each user, follows Rician fading whereas the direct link channel follows Rayleigh fading. Thus, $\mathbf{G}$ and $\mathbf{h}_{r,k}$ are modelled as

$$\mathbf{G} = \mathrm{PL}_1\left(\sqrt{\frac{\kappa}{\kappa+1}}\boldsymbol{\psi}_N(\vartheta)\boldsymbol{\psi}_M(\phi)^H + \sqrt{\frac{1}{\kappa+1}}\tilde{\mathbf{G}}\right)$$

$$\mathbf{h}_{r,k} = \mathrm{PL}_{2,k}\left(\sqrt{\frac{\kappa}{\kappa+1}}\boldsymbol{\psi}_N(\zeta_k) + \sqrt{\frac{1}{\kappa+1}}\tilde{\mathbf{h}}_{r,k}\right),$$

respectively, where $\tilde{\mathbf{G}}$ and $\tilde{\mathbf{h}}_{r,k}$ denote the non-line-of-sight components whose elements follow the distribution $\mathcal{CN}(0,1)$, $\boldsymbol{\psi}$ is the steering vector, κ represents the Rician factor, and ϑ, ϕ and ζ_k the angular values. The system parameters are listed in Table I.

TABLE I
Simulation Parameters

Parameters	Values
BS location	(0m, 0m)
RIS location	(200m, 0m)
Path-loss for $\mathbf{G}$ and $\mathbf{h}_{r,k}$	35.6 + 22.0lg(d)
Path-loss for $\mathbf{h}_{d,k}$	32.6 + 36.7lg(d)
Transmission bandwidth	20MHz
Power spectral density of noise	-174 dBm/Hz
User weight parameters $w_k, \forall k$	1
Iteration layers L	6
PGD steps T	4 or 8
Rician factor κ	10

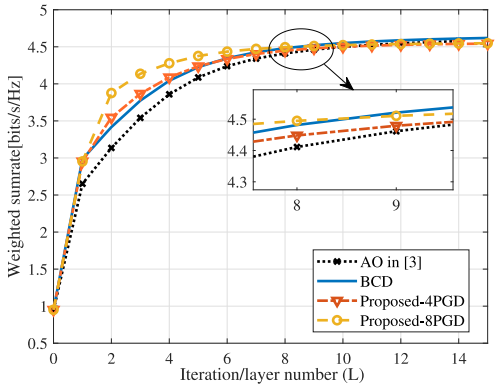


Fig. 3. WSR over iteration number with $N = 80$ and $P_{\max} = 0$ dBm.

Then, we give the NN parameters. The learning rate of PGD operation is initialized to 1, i.e., $\beta_l^{(t)} = 1, \forall l, t$. The Adam optimizer is adopted with a learning rate of 10^{-3} and a minibatch size of 64 channel realizations. There are in total 4000 training batches.

The BCD algorithm and the alternating optimization (AO) algorithm in [3] are also simulated for comparison. Fig. 3 depicts the WSR of our proposed scheme, the AO algorithm, and the BCD algorithm with $N = 80$ and $P_{\max} = 0$ dBm. It is observed that the proposed scheme has fast convergence speed and can achieve WSR performance comparable to the BCD and AO algorithms. We also find that the proposed scheme with 8 PGD steps achieves better performance than that with 4 PGD steps. This is because more trainable parameters leads to more accurate approximation of the transmit beamforming and phase shift solutions.

Fig. 4 shows the relationship between the WSR performance and the transmit power budget of the BS. It can be seen that with the increase of $P_{\max}$, the proposed scheme, as well as the AO and BCD algorithms, can achieve larger WSR. This is because more available power brings more performance gain. Similar to the results in Fig. 3, the performance of the proposed scheme is upper bounded by that of the BCD algorithm, although their performance curves are close to each other.

Fig. 5 depicts the algorithm execution time of the proposed scheme and the AO and BCD algorithms with respect to the number of RIS elements. It is found that the proposed scheme has much less execution time than the AO and BCD algorithms. This is because the AO and BCD algorithms

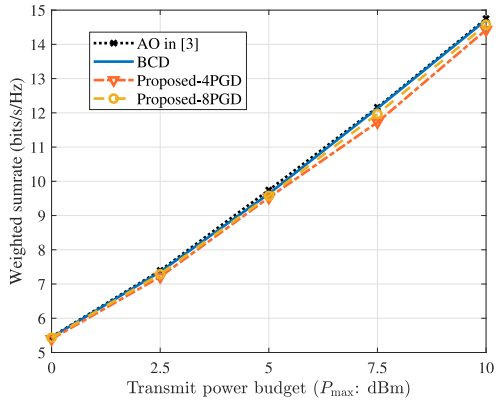


Fig. 4. WSR versus the transmit power budget of the BS $P_{\max}$ with $N = 20$.

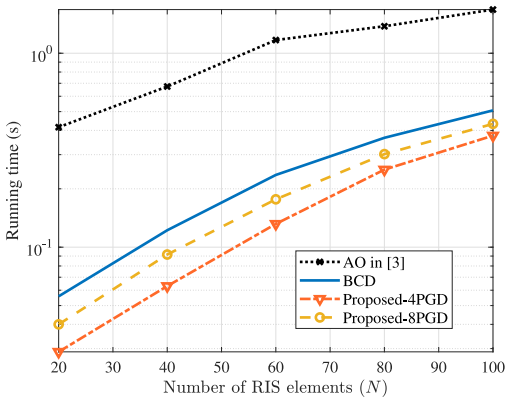


Fig. 5. Algorithm execution time versus the number of RIS elements N.

rely on iterative process whereas the proposed scheme only needs to perform prediction using the trained NN. Based on the results in Figs. 3-5, it can be concluded that the proposed scheme can achieve fast solutions at the cost of a little performance loss. In other words, less execution time is required by the proposed scheme to achieve comparable performance to the AO and BCD algorithms.

VI. Conclusion

This letter focused on the WSR maximization problem by jointly optimizing downlink beamforming of a BS and phase shift of a RIS under the power constraints of the BS. To deal with the non-convex problem, the Lagrangian dual transformation was adopted and the original problem is decomposed into the beamforming optimization subproblem of the BS and the phase shift optimization subproblem of the RIS. An iterative algorithm was proposed. Furthermore, in order to computational complexity and meet realtime requirements, the DUFP algorithm which consists of two modules for updating the downlink beamforming of the BS and phase shift of the RIS. Finally, our numerical results revealed that the proposed scheme can balance between the performance and computational complexity.

It is worth pointing out that the proposed DUFP algorithm can still work in the multi-RIS scenarios where a central RIS-controller exists. However, the decentralized multi-RIS optimization is not well investigated, which deserves further research.

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